

NetPractice (Guidelines)

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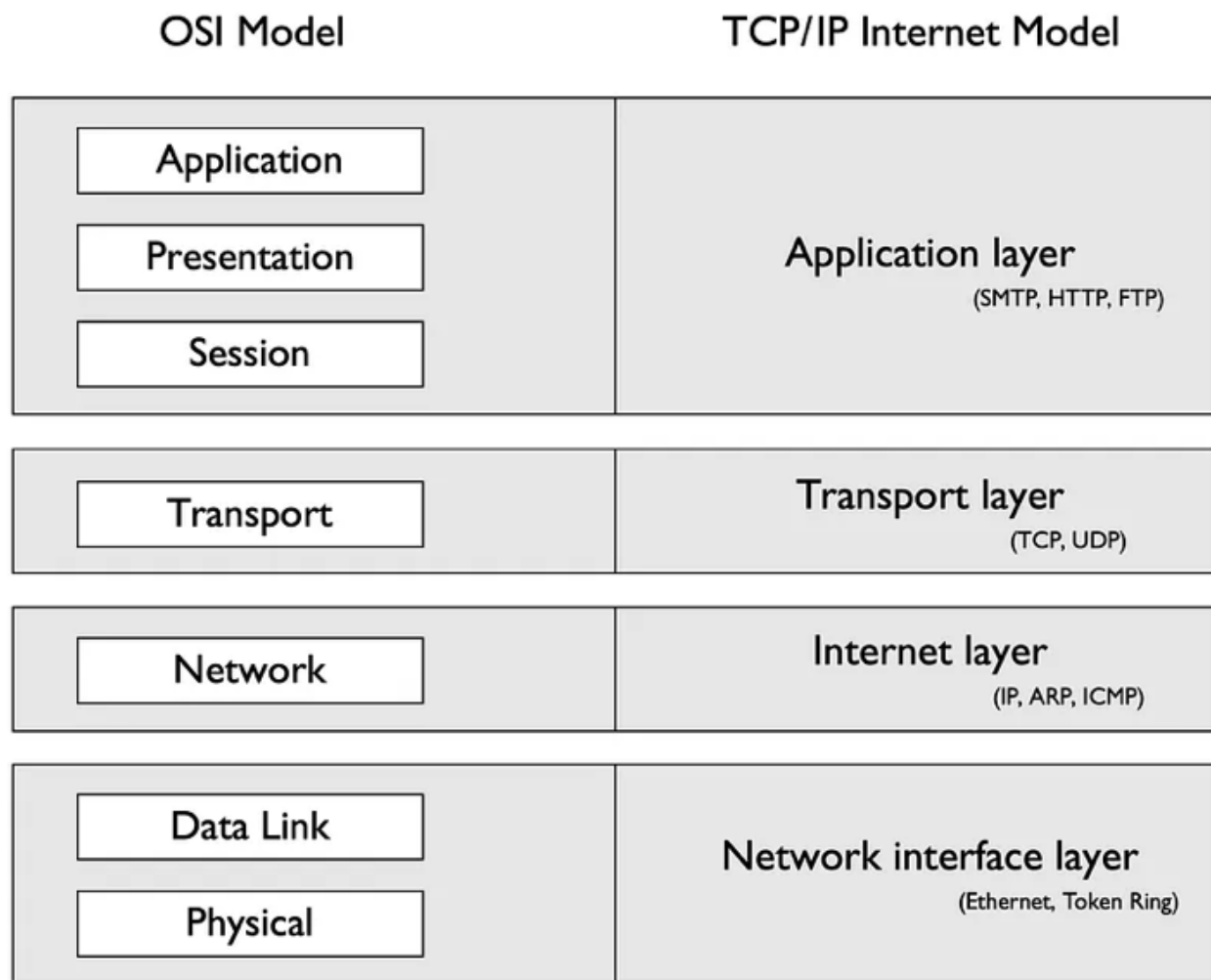
Hello Peers! Hope you're doing good ?!

The most popular protocol in use today is Transmission Control Protocol/Internet Protocol (TCP/IP). The Internet and most company intranets currently use TCP/IP because of its popularity, flexibility, compatibility, and capability to perform in both small and large network implementations.

This article gives you an in-depth understanding of TCP/IP, including the architecture, addressing issues, and a lot more!

What's TCP/IP Protocol :

The TCP/IP Model :



The TCP/IP Internet model versus the OSI model

Contained within the TCP/IP model is a 4-layer model similar in concept to the 7-layer OSI model.

Application layer :

Is responsible for making the network request (sending computer) or servicing the request (receiving computer). For example, when a user submits a request from a web browser, the web browser is responsible for the submission of the request and is running at this layer. When that web request reaches the web server, the web server, running at the application layer, accepts that request. The following are popular application-layer protocols

Transport Layer :

The next layer under the application layer is the transport layer. The transport layer is responsible for both connection-oriented communication (a session is established) and connectionless communication (a session is not established). When the request comes down from the higher (application) layer, a transport protocol is then chosen. The two transport protocols in TCP/IP are the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP).

Internet Layer :

After a transport protocol has been selected, which boils down to whether the communication should be connection-oriented or connectionless, the information is passed to the Internet layer to determine who is responsible for the delivery of the information. There are a few protocols that run at the Internet layer; IP, ICMP, and ARP are examples.

TCP/IP Settings

A TCP/IP protocol requires a little bit of knowledge to configure it properly so as a network manager you are required to know the settings for the:

- IP Address
- Subnet Mask
- Default Gateway

IP Address:

The IP Address is a 32-bit value that uniquely identifies the system on a network (or the Internet), an IP Address looks like this

192.168.1.121

A 4 decimal values separated by (.) each value is made up of 8 bits and 4 decimal values so $8 \times 4 = 32$ -Bit Address.

It's very important to understand that the four octets (We know that each of the decimal values is an octet) in an IP Address are divided into 2 parts :

Host ID

192.168.1.121

Network ID

The diagram shows the IP address 192.168.1.121. A red bracket under the first two octets (192.168) is labeled 'Network ID'. Another red bracket over the last two octets (1.121) is labeled 'Host ID'.

The subnet mask determines the number of bits that make up the network ID and the number of bits that make up the host ID, let's see how this works!

Subnet Mask:

When looking at a subnet mask, if there is a 255 in an octet, then the corresponding octet in the IP Address is part of the network ID, Example :

IP Address 192.168.1.121
Subnet Mask 255.255.255.0



Network ID Host ID

As you see the first 3 octets are reserved for the network ID (This means that any other system on the same network will have the same network ID) and the last for the host ID, The Network ID assigns a unique address to the network itself, while the Host ID uniquely identifies the system on the network.

A good question will be what's the purpose of the subnet masks ??

The reason is that when a system such as 192.168.1.121 with a subnet mask of 255.255.255.0 send a piece of data to 192.168.65.221 the sending system first needs to determine whether the target device exists on the same network or not, it does by comparing the networks IDs :

#	Octet 1	Octet 2	Octet 3	Octet 4
IP Address 1	192	168	1	121
Subnet Mask	255	255	255	0
IP Address 2	192	168	65	221

Identifying 2 systems on different networks using the subnet Masks

so if the network IDs are the same, then both systems exist on the same network and one system can send to the other without the use of a router, if the systems exist on different networks the data will need to be passed to

the router so that the router can send the data to the other network.

Default Gateway:

When your system wants to send data to another system on the network it looks at its *own network ID* and compares that to the *destination's IP address*. If both have the same network ID, the data is sent directly from your system to the destination system. If the two systems are on different networks, your system must pass the data to the router so that the router can send the data to the destination system's router.

The default gateway is the router's IP Address that can send data from your network.

At this point, you have to know that in order to communicate with the internet your system must need to be configured with :

- IP Address
- Subnet Mask
- Default Gateway

and if you need only to communicate with your own network you only need an IP Address and a subnet mask.

IP Addressing:

We're going to talk about the binary representation of an IP Address ;

Understanding Binary:

Let's take our IP Address 192.168.1.121, the binary representation looks like this :

11000000 10101000 00000001 01111001

notice that there are 4 sets of 8 bits (1 or 0) which makes up 32-bits of an IP Address

Bit	8	7	6	5	4	3	2	1
Value	128	64	32	16	8	4	2	1

In the table above as you see the value far right it's 1 the second 2 the third 4 and the values keep doubling with each additional bit.

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So to calculate the binary value of an octet with a decimal number such as 234 you need to calculate the combination of numbers that will add up to the number 255 as demonstrated in the table below :

Bit	8	7	6	5	4	3	2	1
Value	128	64	32	16	8	4	2	1
State (1 or 0)	1	1	1	0	1	0	1	0

$$234 = 128 + 64 + 32 + 8 + 2$$

Notice that a value of $128 + 64 + 32 + 8 + 2$ which equals 234. The remaining bits will take an "0" off state, which means they are not included in the

calculation. A bit that has an on state takes a “1” and an off state takes a “0.” So the combination of 8 bits to make the number 234 would be 11101010.

Address Classes:

Every IP Address belongs to a distinct address class. The class to which the IP Address belongs initially determines the network ID and host ID portion of the address. The different class addresses are named class A, class B, class C, class D, and class E:

Class A Addresses:

Class A addresses have an IP address in which the first octet is *between 1 and 126*. Class A addresses also have a default subnet mask of *255.0.0.0*

This means that the first octet is reserved for the network ID and the last 3 for the Host ID, a class A address supports 16,777,216 hosts on the network (256 x 256 x 256) 2, actually there are only 16,777,214 valid addresses to use on systems because there's on each IP Network you are not allowed to assign to systems because they are reserved.

These are the addresses with all host bits set to 0s (the network ID) and all host bits set to 1s (the broadcast address).

So with a class A address, you will not be able to assign n.0.0.0 or n.255.255.255 (where n is your network ID) to any hosts on the network.

Class B Addresses:

Class B addresses have an IP address in which the value of the first octet is *between 128 and 191*. Class B addresses have a default subnet mask of *255.255.0.0*.

which means that the first two octets are the network ID and the last two octets are the host ID portion of the address. This means that we can have 65,536 hosts (256 × 256) on the network. but wait! Don't forget to take off the

two *illegal addresses*, so that gives us 65,534 addresses that can be assigned to hosts on the network.

Illegal Addresses: *There are a few IP addresses that are illegal to assign to hosts on the network.*

Actually, an address that starts with 127 is a class A address as well, but you are not allowed to use any address that starts with 127, because it is reserved for the loopback address

Loopback Address : The loopback address is used to refer to the local system, also known as the localhost.

Class C Addresses:

Class C addresses have an IP address in which the value of the first octet is *between 192 and 223*. In addition, class C addresses have a default subnet mask of **255.255.255.0**.

means that the first three octets are the network ID and the last octet is the host ID. Having only one octet as the host ID means that a class C address can support only 254 hosts (256-2) on the network.

let's not talk about D and E ...

a quick summary of IP Addresses classes :

	First Octet Value	Subnet Mask	# of Hosts per Network
Class A	1–126	255.0.0.0	16,777,214
Class B	128–191	255.255.0.0	65,534
Class C	192–223	255.255.255.0	254

Address Classes

Private Addresses:

A private address is an address that can be assigned to a system but cannot be used for any kind of Internet connectivity. The private addresses are non-routable addresses, so any system using them will not be able to function off the network.

The following are the three address ranges that are private:

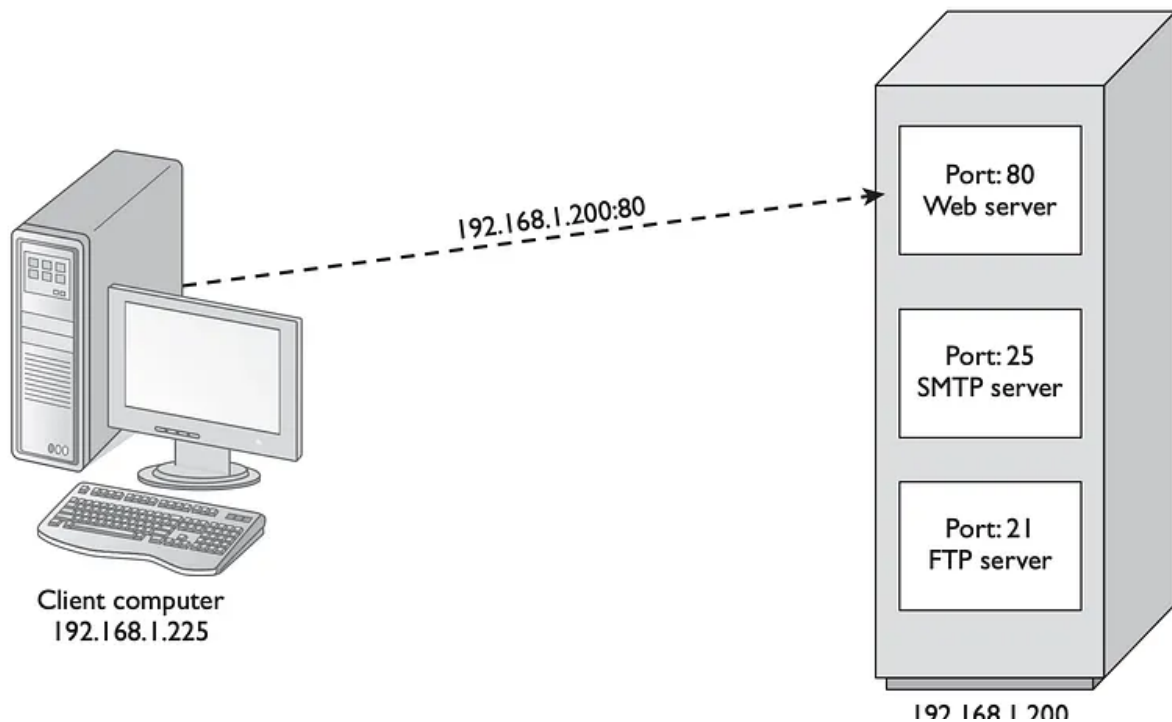
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

Private IP Address Ranges

TCP/IP Ports :

An application or process uses a TCP/IP port as an identifier for that application running on a system. When you send data from one computer to another, you send data to the port used by that application.

For example, when you type the IP address of a website in your web browser, the web browser connects to the web server running at that system by sending data to port 80 (Simple isn't)

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Now that you have a handle on binary and TCP/IP addressing, you will learn in this chapter how to take one network address range and break it into multiple networks ranges called subnets.

Subnetting is one of those topics that most IT professionals tend to avoid because of how tedious it is to sit down and calculate the required information to configure your newly created subnets.

What's the purpose of subnetting?

The purpose of subnetting is to *take one address range* that you have and break it down into *multiple addresses ranges* so that you can assign each address range to a separate network (subnet) in your internetwork (network made up of multiple networks).

And what if we didn't subnet our network?

If you do not subnet a network range that is divided into multiple physical network segments, the IP protocol will “logically” think that *all systems are on the same network*, and it will not try to route the data across the router separating the network segments. As a result, you will be able to communicate with hosts on your segment but not the other segments.



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